SE Lab 3
Architectural Pattern Selection

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Vaccination Cohort & Dose Scheduling System

Architecture Selection

We chose Layered Architecture for the Vaccination Cohort & Dose Scheduling System.

Architectural Choice

A three-tier Layered Architecture : Presentation Layer, Business Layer, and Data Layer was selected. The Presentation Layer exposes a single User Interface Component used by both citizens and vaccination officers. The Business Layer contains four components: the Registration & Eligibility Manager, Slot Booking Manager, Vaccination Record Manager, and QR Certificate Manager. The Data Layer holds the Vaccination Database Component that stores registrations, slots, dose history, and certificates.

Two Reasons for This Choice

1. Centralised rule enforcement: Rules such as the 28-day minimum dose interval (FR-001) and prevention of slot double-booking (NFR-002) must be applied consistently across every citizen and officer interaction. Placing this logic in one Business Layer rather than scattering it across UI screens or database triggers means each rule is implemented once and reused by every component that needs it.

2. Right-sized for the scope: The system serves three actor types (Citizen Registrant, Vaccination Officer, QR Verification Service) through a small, stable set of features. This does not justify the operational overhead of independently deployed microservices; a layered structure gives sufficient modularity while keeping deployment, testing, and debugging simple for a small team.

Security Advantage

The Presentation Layer never talks to the Data Layer directly. Every request must pass through the Business Layer first. This means authentication, eligibility checks, and slot-availability checks are always enforced before any read or write reaches the Vaccination Database Component, closing off a path where a compromised or buggy UI could otherwise tamper with vaccination records directly.

Performance Benefit

Because QR signature verification (NFR-001, target < 150 ms) is handled entirely inside the QR Certificate Manager in the Business Layer, the QR Verification Service only needs one call across the system boundary. There is no extra network hop to a separate verification microservice, which keeps the verification path short and predictable under peak load.